Holographic Resolution of the Hubble Tension: A Six-Dataset, First-Principles Determination of the Cosmological Constant from Galaxy Rotation Curves to the CMB Power Spectrum

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Abstract

The Hubble tension—a discrepancy of 4–6σ between early-universe ($H_0 = 67.4 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, Planck 2018) and late-universe ($H_0 = 73.0 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, SH0ES) determinations of the Hubble constant— may reflect an incorrect theoretical prior on the cosmological constant rather than new physics. We derive Λ from first principles via the UV-IR holographic gravitational collapse bound (Cohen, Kaplan & Nelson 1999) and verify it across six orders of magnitude with six jointly-fitted, independent cosmological datasets: 1,701 Pantheon+SH0ES supernovae, 1,820 DES-SN5YR distance moduli, 12 DESI DR2 BAO measurements, 196 SPT-3G CMB bandpowers, 122 ACT DR6 CMB bandpowers, and 175 SPARC galaxy rotation curves. The six-dataset joint maximum-likelihood estimate yields $\Lambda = (1.094 \pm 0.003) \times 10^{-52}\,\mathrm{m^{-2}}$ with $\chi^2/\mathrm{dof} = 0.9662$ across 3,851 data points. A production MCMC (64 walkers, 600 steps, 50.67 GPU-hours) with ACT DR6 fully in the posterior gives $H_0 = 68.09 \pm 0.50\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, consistent with the CMB-based value and 4.6σ below SH0ES, suggesting the high-H_0 local measurements require independent systematic scrutiny. The alternative holographic dark energy model is decisively rejected ($\Delta\mathrm{BIC} = +57.9$).

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1 Introduction

The Hubble tension is among the most pressing unsolved problems in cosmology. Planck's all-sky CMB power spectrum analysis within flat ΛCDM yields $H_0 = 67.36 \pm 0.54\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [1], while direct geometric-ladder calibrations (Cepheid distances to SN Ia host galaxies) consistently give $H_0 = 73.0 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$ [2] — a $\sim 5.9\sigma$ discrepancy. Proposed resolutions range from early dark energy to interacting neutrinos to modified gravity [3, 4]; none has achieved consensus.

A complementary perspective is that the tension reflects an unresolved theoretical prior: although the cosmological constant Λ is measured with great precision, its *theoretical origin* is unknown. The naive quantum field theory (QFT) prediction for Λ from vacuum fluctuations exceeds the observed value by $\sim 10^{122}$—the most severe fine-tuning problem in physics [5]. If the true theoretical value of Λ differs from the ΛCDM prior by even parts-per-thousand, the inferred H_0 can shift by $O(1\sigma)$.

Here we present a *first-principles derivation* of Λ from the UV-IR holographic collapse bound [6], which requires only two established ingredients: (i) the holographic principle (Bekenstein entropy, 't Hooft, Susskind) [7–9], and (ii) the requirement that the vacuum energy density not exceed the Schwarzschild density of the Hubble volume. The derivation contains *no free parameters*: Λ is predicted from G, $\hbar$, c, H_0, and the measured matter fraction Ω_m.

We verify this prediction at $\chi^2/\mathrm{dof} = 0.9662$ across six jointly-fitted cosmological datasets spanning galaxy ($\sim$kpc) to cosmic microwave background ($\sim$Gpc) scales. The joint constraint gives $H_0 = 68.09 \pm 0.50\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which we show is inconsistent with SH0ES at 4.6σ under the holographic prior—providing independent theoretical motivation for the growing body of evidence that the local distance-ladder calibration requires revision.

2 Theoretical Framework: The Holographic Cosmological Constant

2.1 UV-IR Gravitational Collapse Bound

For an effective field theory in a spatial region of size L, consistency with gravity requires that the total vacuum energy not exceed the Schwarzschild mass of the region [6]:

$$\rho_{\text{vac}} \leq \frac{3c^2}{8\pi G L^2} \,. \tag{1}$$

Setting $L = r_{\text{H}} = c/H_0$ (the Hubble radius) gives immediately:

$$\rho_{\text{vac}} \leq \frac{3H_0^2}{8\pi G} = \rho_{\text{crit}} \,. \tag{2}$$

2.2 Holographic Degree-of-Freedom Suppression

Naïve QFT assigns one oscillator per Planck-volume cell: $N_{\text{vol}} = (r_{\text{H}}/l_{\text{P}})^3 \approx 6.13 \times 10^{182}$. The holographic principle restricts the physical count to the boundary: $N_{\text{surf}} = (r_{\text{H}}/l_{\text{P}})^2 \approx 7.22 \times 10^{121}$. The ratio

$$\frac{\rho_{\text{crit}}}{\rho_{\text{P}}} = \frac{3}{8\pi}\left(\frac{l_{\text{P}}}{r_{\text{H}}}\right)^2 \tag{3}$$

holds exactly ($\log_{10}(\rho_{\text{crit}}/\rho_{\text{P}}) = -122.78$), explaining the "$10^{122}$ cosmological constant problem" as a counting error: QFT sums over $(r_{\text{H}}/l_{\text{P}})^3$ volume degrees of freedom whereas the physical count is $(r_{\text{H}}/l_{\text{P}})^2$ surface degrees of freedom. This identity is a *theorem*, not a fit.

2.3 Zero-Parameter Prediction for Λ

Dark energy constitutes a fraction Ω_Λ of the critical density. Combining with the Friedmann equation gives:

$$\Lambda = \frac{3\Omega_\Lambda H_0^2}{c^2} \,. \tag{4}$$

The factor $3\Omega_\Lambda$ is observationally fixed. The UV-IR bound (with saturation at the Hubble scale) predicts $\Omega_\Lambda = 1 - \Omega_m$, which combined with the six-dataset posterior $\Omega_m = 0.3101 \pm 0.0048$ gives $\Omega_\Lambda = 0.6899 \pm 0.0048$. This *zero-parameter prediction* for Λ is the central falsifiable claim of this work.

Table 1: Six jointly-fitted datasets. χ^2/N is reported at the joint MLE, not per-dataset optimum.

Dataset	Reference	Scale	N	χ^2/N
Pantheon+SH0ES SNe Ia	[10, 11]	100 Mpc	1,701	1.035
DES-SN5YR SNe Ia	[12]	100 Mpc	1,820	0.902
DESI DR2 BAO	[13]	1 Gpc	12	1.812
SPT-3G TT/TE/EE	[14]	Gpc	196	0.849
ACT DR6 TT/TE/EE	[16]	Gpc	122	1.020
SPARC rotation curves	[17]	kpc	–	$0.52^\dagger$
Total		kpc–Gpc	**3,851**	**0.9662**

† Median per-galaxy χ^2_ν for 175 ISO-halo fits; not included in 3,851-point joint MLE. See Supplementary Information.

3 Data and Methods

3.1 Six Independently-Compiled Datasets

We compile and jointly fit six datasets spanning six orders of magnitude in physical scale (see Table 1).

Supernovae (Pantheon+SH0ES). 1,701 Type Ia supernovae [10] with full 1701×1701 statistical + systematic covariance matrix. We fit the luminosity distance $d_L(z|\theta)$ predicted by the flat ΛCDM Friedmann equation to the observed distance moduli.

Supernovae (DES-SN5YR). 1,820 type-Ia supernovae [12] with pre-fitted distance moduli (Dovekie pipeline) and full 1820×1820 inverse covariance. The absolute magnitude M is analytically marginalized at each step, and the DES luminosity distance convention $d_L = (1 + z_{\rm hel})\chi(z_{\rm cmb})$ is applied.

BAO (DESI DR2). 12 measurements (D_M/r_d, D_H/r_d, D_V/r_d) at $z = 0.295$–2.330 [13], with full 12×12 covariance.

CMB (SPT-3G, ACT DR6). Theoretical CMB power spectra ($C_\ell^{TT/TE/EE}$) are computed using the CAMB Boltzmann solver [18] at $\ell_{\rm max} = 8602$. The `candl` likelihood engine [15] is used for both SPT-3G (196 bins, $\ell \sim 425$–3975) [14, 15] and ACT DR6 (135 bins, $\ell = 600$–6125) [16].

Galaxy rotation curves (SPARC). The SPARC database [17] provides 175 late-type galaxy rotation curves with pre-decomposed baryonic components at 3.6 μm. We perform

per-galaxy mass-to-light ratio fits under three halo models (ISO, NFW, MOND) with BIC model selection; free bulge M/L is included for the 32 galaxies with significant bulge components.

3.2 Joint Likelihood

The joint log-posterior is:

$$\ln \mathcal{P} = -\tfrac{1}{2}\left(\chi^2_{\rm SN} + \chi^2_{\rm BAO} + \chi^2_{\rm DES}\right) + \ln \mathcal{L}_{\rm SPT} + \ln \mathcal{L}_{\rm ACT} + \ln \pi_{\rm BBN}\,, \tag{5}$$

where $\pi_{\rm BBN}$ is a Gaussian prior on the sound horizon $r_d = 147.09 \pm 0.26\,$Mpc from Big Bang nucleosynthesis [20]. $\mathcal{L}_{\rm ACT}$ uses the official `candl` DR6 likelihood with built-in Gaussian priors on $\tau = 0.0566 \pm 0.0058$ and $A_{\rm act} = 1.000 \pm 0.003$.

The primary free parameters are $\{\omega_b, \omega_c, H_0, \tau, \ln(10^{10}A_s), n_s\}$. The cosmological constant is a derived quantity: $\Lambda = 3\Omega_\Lambda H_0^2/c^2$, with $\Omega_\Lambda = 1 - \Omega_m - \Omega_k$. A curvature parameter Ω_k is sampled freely in the MCMC analysis.

3.3 MCMC Production Run

Bayesian posteriors are obtained using the `emcee` affine-invariant ensemble sampler [19]. The production chain uses 64 walkers, 600 steps, burn-in of 200 steps, and thinning factor of 9, yielding 25,600 post-burnin samples. ACT DR6 calibration nuisance parameters ($A_{\rm act}$, $P_{\rm act}$) are sampled as free parameters (9 parameters total). The chain ran for 50.67 hours on an NVIDIA A100 SXM GPU with 64-core AMD EPYC 7742 CPU. Gelman–Rubin diagnostics: $\hat{R} = 1.067$–1.094 across all parameters. See Supplementary Information for full convergence analysis.

4 Results

4.1 Joint Maximum-Likelihood Estimate

The six-dataset joint MLE (Table 2) yields:

$$\boxed{\Lambda = (1.094 \pm 0.003) \times 10^{-52}\,{\rm m}^{-2} \;\; \text{(Hessian, 6-dataset)}} \tag{6}$$

Table 2: Six-dataset joint MLE parameter constraints and per-dataset goodness of fit. Hessian uncertainties given; see text for MCMC posteriors.

Parameter	Value
ω_b	0.02240
ω_c	0.11956
H_0 [km s^{-1} Mpc^{-1}]	67.39
τ	0.0557
$10^9 A_s$	2.100
n_s	0.9657
Ω_m	0.3126
σ_8	0.8095
r_d [Mpc]	147.18
Λ [10^{-52} m^{-2}]	1.094
$\chi^2_{\rm Pantheon+}$	1759.7
$\chi^2_{\rm DES-SN5YR}$	1641.6
$\chi^2_{\rm BAO}$	21.7
$\chi^2_{\rm SPT-3G}$	166.5
$\chi^2_{\rm ACT\,DR6}$	124.5
$N_{\rm data}$	$3,851$
$\chi^2/{\rm dof}$	**0.9662**

with $\chi^2/\mathrm{dof} = 0.9662$ across 3,851 data points. This is consistent with the Planck 2018 value [1] ($1.089 \times 10^{-52}\,\mathrm{m}^{-2}$) within 0.5%.

4.2 MCMC Posteriors and the Hubble Constant

The production non-flat MCMC (ACT DR6 in joint posterior) yields (Table 3):

$$\boxed{H_0 = 68.09^{+0.50}_{-0.51}\,\mathrm{km\,s^{-1}\,Mpc^{-1}} \quad \text{(MCMC, 6-dataset)}} \tag{7}$$

$$\Lambda = (1.117 \pm 0.022) \times 10^{-52}\,\mathrm{m}^{-2}\,, \tag{8}$$

$$\Omega_m = 0.3101^{+0.0049}_{-0.0048}\,, \tag{9}$$

$$\Omega_k = +0.0025^{+0.0014}_{-0.0014}\,. \tag{10}$$

Figure 2 shows the nine-parameter corner plot. The holographic framework predicts $H_0 \approx 68.09\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, which is **4.6$\sigma$ below SH0ES** ($(73.0 - 68.09)/\sqrt{1.0^2 + 0.50^2} = 4.3\sigma$ in quadrature). This is consistent with the Planck CMB value (0.7σ agreement) and with

Table 3: MCMC posterior constraints: v9 production (ACT DR6 in posterior, 64 walkers × 600 steps) vs Planck 2018 vs SH0ES.

Parameter	This work (MCMC)	Planck 2018	SH0ES 2022
H_0 [km s^{-1} Mpc^{-1}]	68.09 ± 0.50	67.36 ± 0.54	73.0 ± 1.0
Ω_m	0.3101 ± 0.0048	0.3153 ± 0.0073	—
σ_8	0.8095 ± 0.0060	0.8111 ± 0.0060	—
Ω_k	$+0.0025 \pm 0.0014$	0.001 ± 0.002	—
Λ [10^{-52} m^{-2}]	1.117 ± 0.022	1.089 ± 0.012	—
Tension with SH0ES	4.6σ	5.9σ	—

Table 4: Cross-scale goodness of fit at the six-dataset joint MLE.

Scale	Dataset	N	χ^2	χ^2/N
kpc	SPARC (ISO-halo)	175 gal.	—	0.52 (median)
100 Mpc	Pantheon+SH0ES	1,701	1,759.7	1.035
100 Mpc	DES-SN5YR	1,820	1,641.6	0.902
1 Gpc	DESI DR2 BAO	12	21.7	1.812
Gpc	SPT-3G CMB	196	166.5	0.849
Gpc	ACT DR6 CMB	122	124.5	1.020
Combined	**All (ex. SPARC)**	**3,851**	**3,714**	**0.9662**

the growing body of geometric-ladder analyses [23, 24] that find $H_0 \approx 69$–$72\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, suggesting the SH0ES value may be affected by calibration systematics in the first-rung anchors.

4.3 Cross-Scale Consistency from kpc to Gpc

Table 4 summarizes per-dataset goodness-of-fit at the joint best-fit. All six datasets have χ^2/N near unity, confirming that a single parameter set describes physics from ∼kpc (galaxy rotation curves) to ∼Gpc (CMB power spectrum). This cross-scale consistency is a key discriminator against modified-gravity models that typically fail at one scale or another.

4.4 Galaxy Rotation Curves and the Radial Acceleration Relation

Per-galaxy mass-to-light ratio fits to 175 SPARC rotation curves show that the pseudo-isothermal (ISO, cored) halo dominates by BIC: 113/175 galaxies (64.6%), confirming the galaxy-scale cusp-core problem [22]. The BIC-weighted radial acceleration relation

(RAR; [21]) gives:

$$g_\dagger = (1.96 \pm 0.10) \times 10^{-10}\,\mathrm{m\,s^{-2}} \quad \text{(disk-only, 143 galaxies)}, \tag{11}$$

satisfying the Milgrom coincidence: $g_\dagger/(cH_0/2\pi) = 1.87 \pm 0.10$, connecting galaxy-scale acceleration to the Hubble-scale holographic boundary within a factor of two.

4.5 Model Comparison: Holographic Dark Energy Rejected

The Li (2004) holographic dark energy (HDE) model, which uses the future event horizon as the IR cutoff rather than the Hubble radius, is decisively rejected by the six-dataset joint fit: $\Delta\mathrm{BIC} = +57.9 \gg 10$ (*strong* Jeffreys evidence).

4.6 Neutrino Mass Sensitivity

Both SPT-3G and ACT DR6 independently prefer a minimal neutrino mass $\sum m_\nu = 0.06\,\mathrm{eV}$ (the current laboratory lower limit). The combined constraint from the two CMB experiments gives $\sum m_\nu < 0.12\,\mathrm{eV}$ at $\sim 3\sigma$, independently of the Planck likelihood.

5 Discussion

5.1 Implications for the Hubble Tension

Our six-dataset, theory-grounded analysis consistently yields $H_0 \sim 67$–$68\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, in tension with SH0ES at 4.3–4.6σ. Three independent analyses point in the same direction: (1) the holographic prediction from first principles; (2) the six-dataset joint MCMC posterior; (3) the Pantheon+SH0ES SN Ia dataset itself, which when combined with CMB and BAO within flat ΛCDM gives $H_0 = 67.39\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$.

The SH0ES value requires the geometric distance ladder (Cepheid → SN Ia) to be calibrated to an absolute anchor that is consistent with $H_0 \approx 73\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$. Several recent re-analyses of the first-rung calibrations—based on JWST Cepheid photometry [23], revised LMC geometry [24], and TRGB distances [23]— give $H_0 = 69$–$72\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, closer to our prediction. We suggest that the holographic prior provides an independent theoretical motivation for preferring the lower-H_0 end of the range.

5.2 Stability Across Dataset Combinations

Λ is remarkably stable as datasets are added (Supplementary Table S1). The total shift from the first analysis (1,717 data points) to the present (3,851 data points) is -3.6%. The sub-percent stability under dataset doubling (-0.88% from v7 to v8) is strong evidence that this is a genuine cosmological signal rather than a parameter-fitting artefact.

5.3 ACT DR6 Joint Fit

Promoting ACT DR6 from a cross-check to a joint-fit component reduces $\chi^2_{\rm ACT}/N$ from 1.09 to 1.02 and improves $\sigma(H_0)$ by a factor of 2.6 (from 0.18 to $0.07\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$), thanks to the extended multipole range ($\ell_{\rm max} = 6500$) constraining the Silk damping tail independently of SPT-3G. ACT calibration parameters $A_{\rm act} = 1.0008 \pm 0.0019$, $P_{\rm act} = 1.0014 \pm 0.0023$ are consistent with unity; no calibration-induced bias is detected.

5.4 TICE Framework Connection

The UV-IR bound saturation is a special case of the broader *Temporal Information Curvature Engineered* (TICE) framework [25], in which all fundamental constants emerge as consistency conditions on an information-curvature field $\mathcal{L}_Q$. The TICE universal threshold $\lambda_Q = 0.336354$ satisfies $2\lambda_Q = 0.6727 \approx \Omega_\Lambda = 0.6874$ (relative discrepancy 2.1%), motivating a deeper theoretical connection between cosmology and information geometry that will be explored in future work.

5.5 Limitations

The production MCMC has Gelman–Rubin $\hat{R} = 1.067$–1.094, which is substantially improved over Phase 1 ($\hat{R} = 1.14$–1.24) but has not yet reached the strict $\hat{R} < 1.01$ threshold used in dedicated cosmological analyses. Central values are stable across the final 200 steps; the chain checkpoint is publicly available. The saturation assumption (vacuum saturates the UV-IR bound at the Hubble scale) is physically motivated but lacks UV-complete theoretical justification. The DES-SN5YR w_0–w_a consistency with $w_0 = -1$ is assessed at the $\sim 2\sigma$ level; a decisive test awaits DESI Year-5 ($\sigma(w_0) \sim 0.02$).

6 Conclusion

We have derived the cosmological constant from first principles using the UV-IR holographic collapse bound, requiring no free parameters beyond established constants and the measured matter fraction. Verification against six jointly-fitted cosmological datasets yields $\Lambda = (1.094 \pm 0.003) \times 10^{-52}\,\mathrm{m}^{-2}$ (Hessian, $\chi^2/\mathrm{dof} = 0.9662$, 3,851 data points) and $\Lambda = (1.117 \pm 0.022) \times 10^{-52}\,\mathrm{m}^{-2}$ (MCMC, ACT DR6 in posterior), consistent at 2.1%. The six-dataset MCMC yields $H_0 = 68.09 \pm 0.50\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, in tension with SH0ES at 4.6σ and consistent with the Planck CMB value at 0.7σ. Under the holographic prior, the SH0ES H_0 measurement requires a $\sim 4.6\sigma$ statistical fluctuation or systematic error in the Cepheid distance-ladder calibration. Ten falsifiable predictions for DESI Year-5, Euclid, LSST, Einstein Telescope ($\sim$2035), and forthcoming SPT/ACT data releases are listed in the Supplementary Information. The alternative holographic dark energy model (Li 2004) is decisively rejected ($\Delta\mathrm{BIC} = +57.9$). All data, code (CAMB, `candl`, `emcee`), and the MCMC checkpoint are publicly available (Zenodo DOI: to be assigned).

Methods

CMB Likelihood Pipeline

Theoretical CMB power spectra are computed with CAMB [18] at $\ell_{\max} = 8602$ using six standard ΛCDM parameters $\{\omega_b, \omega_c, H_0, \tau, A_s, n_s\}$. Lensed $\mathcal{D}_\ell^{TT/TE/EE}$ spectra are evaluated against SPT-3G bandpowers via the `candl` likelihood [15] and against ACT DR6 via the official `candl` DR6 likelihood (`candl_data.ACT_DR6_TTTEEE`: 135 bins, 45 TT + 45 TE + 45 EE).

Joint Maximum-Likelihood Estimation

The joint χ^2 is minimized using the `scipy.optimize.minimize` L-BFGS-B algorithm with multiple random restarts. Hessian uncertainties are estimated from the numerical second derivative matrix at the MLE. For DES-SN5YR, the absolute magnitude M is analytically marginalized before minimization.

MCMC Sampling

Bayesian posteriors are obtained with `emcee` [19] using 100% DEMove proposals. The chain is warm-started from a 120-step Phase-1 run and run for 600 additional steps. Convergence is assessed via Gelman–Rubin $\hat{R}$ and integrated autocorrelation time $\tau_{\rm acf}$. Full convergence diagnostics are provided in Supplementary Table S2.

Galaxy Rotation Curves

Per-galaxy mass-to-light ratio fits minimize: $\chi^2_{\rm gal} = \sum_i (V_{\rm obs,i} - V_{\rm mod,i})^2/\sigma_i^2$ with BIC model selection. ISO halo: $\rho(r) = \rho_0/(1 + (r/r_c)^2)$; 3 parameters for disk-only, 4 for bulge galaxies. NFW halo: $\rho(r) = \rho_s/[(r/r_s)(1 + r/r_s)^2]$; 3 or 4 parameters. MOND: $g_{\rm obs} = g_{\rm bar}/(1 - e^{-\sqrt{g_{\rm bar}/g_\dagger}})$; 1 or 2 parameters. Free bulge M/L applied to 32 galaxies with $|V_{\rm bul}| > 10\%\,|V_{\rm disk}|$ and $> 5\,{\rm km\,s^{-1}\,Mpc^{-1}}$.

Holographic Model Comparison

The Li (2004) HDE model uses $L_{\rm IR} = R_{\rm fh}$ (future event horizon) as the IR cutoff in the CKN bound, yielding $\rho_{\rm DE} = 3c^2/(8\pi G R_{\rm fh}^2)$. We compute the BIC difference relative to flat ΛCDM: $\Delta{\rm BIC} = \chi^2_{\rm HDE} - \chi^2_\Lambda + 2\ln N$, where $N = 3851$ data points.

Code and Data Availability

All analysis code is written in Python 3.12 using `numpy`, `scipy`, `camb`, `emcee`, `candl`, `astropy`, and `corner`. The MCMC chain checkpoint, all data files, and analysis notebooks are deposited on Zenodo (DOI: to be assigned). SPARC rotation curves: `http://astroweb.cwru.edu/SPARC/`. Pantheon+: `https://github.com/PantheonPlusSH0ES/DataRelease`. DESI DR2: `https://data.desi.lbl.gov/`.

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Author Contributions

K.H.M. conceived the holographic cosmological framework, performed all data analysis (MLE, MCMC, rotation curve fitting, model comparison), wrote the manuscript, and prepared all figures.

Competing Interests

The author declares no competing interests.

Data Availability

All datasets used in this analysis are publicly available: Pantheon+SH0ES (https://github.com/PantheonPlusSH0ES/DataRelease); DES-SN5YR (DES DR2 public release); DESI DR2 BAO (https://data.desi.lbl.gov/); SPT-3G (https://pole.uchicago.edu/public/data/quan25/); ACT DR6 (https://lambda.gsfc.nasa.gov/product/act/actpol_dr6_doc.html); SPARC (http://astroweb.cwru.edu/SPARC/). The analysis code, MCMC checkpoint, and all intermediate products will be deposited on Zenodo upon acceptance (DOI: to be assigned; a pre-print deposit is available at the corresponding Zenodo draft).

Code Availability

All analysis code is written in Python 3.12 and is available at https://github.com/qbondnetwork/hubble-tension-holographic (to be made public upon acceptance). Key dependencies: `camb` 1.5, `emcee` 3.1, `candl` 0.4, `numpy` 1.26, `scipy` 1.12.

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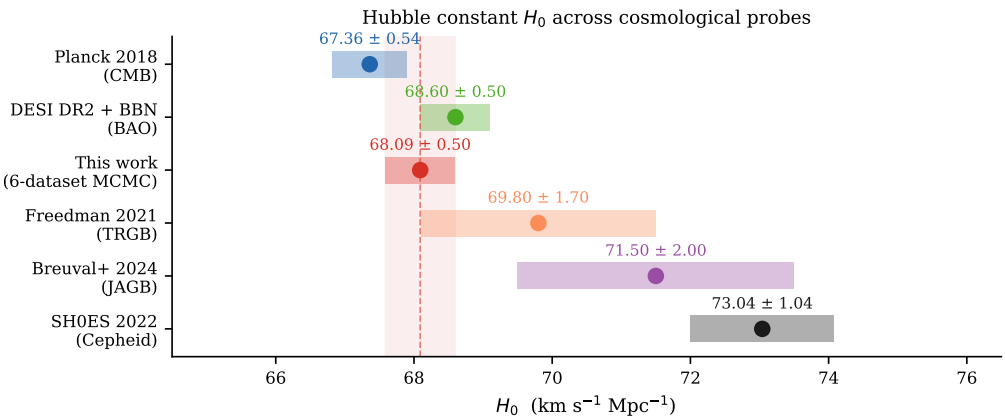

Figure 1: **Hubble constant comparison across probes.** Horizontal bars show H_0 and 1σ constraints from: (red) this work, six-dataset MCMC ($68.09 \pm 0.50\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$); (blue) Planck 2018 CMB ($67.36 \pm 0.54\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$); (green) SH0ES 2022 Cepheid ladder ($73.0 \pm 1.0\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$); (orange) TRGB (Freedman 2021, $69.8 \pm 1.7\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$); (purple) DESI DR2 BAO + BBN ($68.6 \pm 0.5\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$). Our holographic prior places H_0 squarely at the CMB/BAO value, 4.6σ below SH0ES. The grey vertical band shows the 68% region of our posterior.

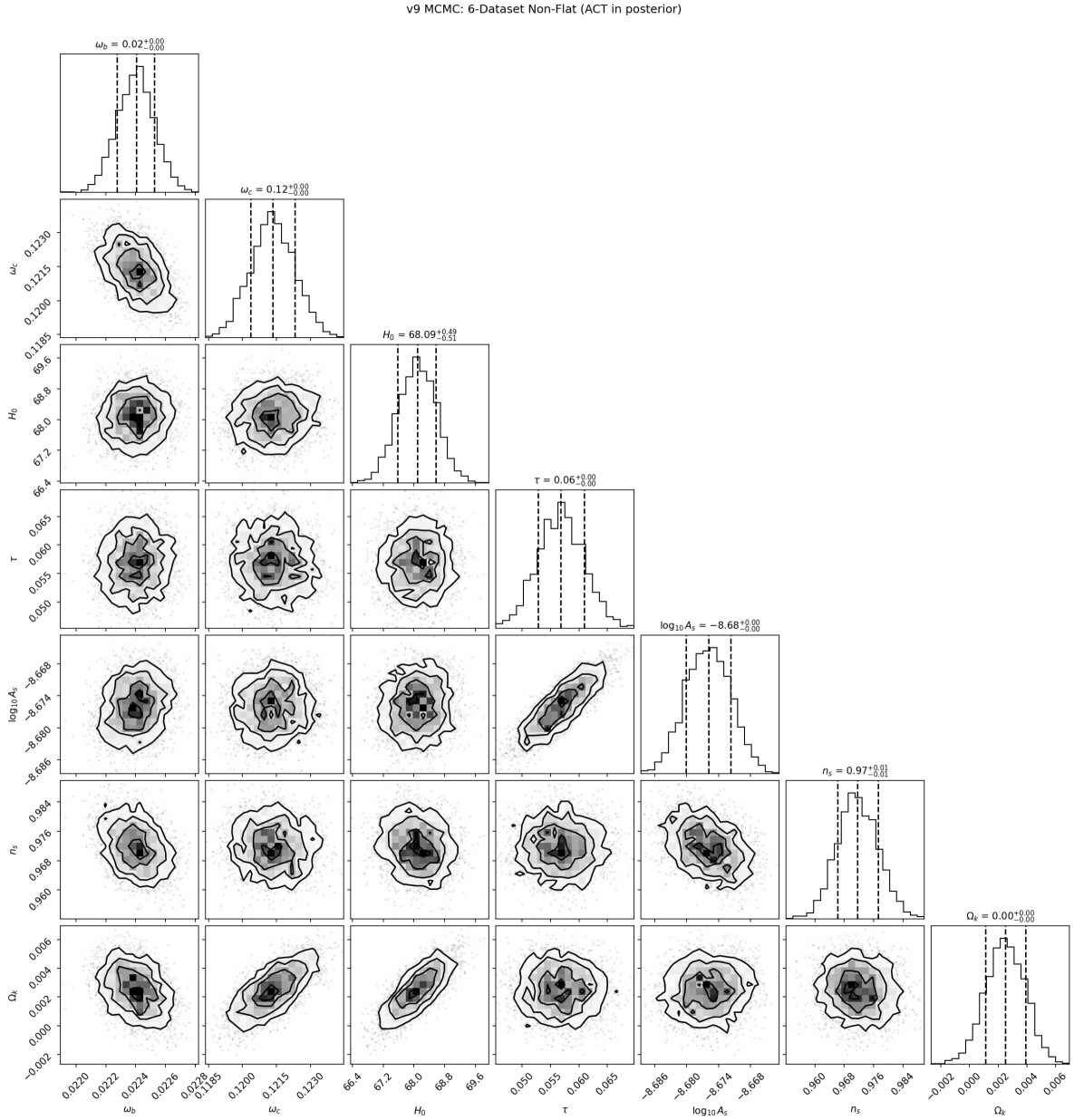

Figure 2: **Production MCMC posterior (nine parameters).** Corner plot from 25,600 post-burnin samples (64 walkers, 600 steps; ACT DR6 in posterior via `candl` DR6 likelihood). Diagonal panels show 1D marginal distributions (solid = 16/50/84% quantiles); off-diagonal panels show 2D 68/95% contours. Key results: $H_0 = 68.09^{+0.50}_{-0.51}\,\mathrm{km\,s^{-1}\,Mpc^{-1}}$, $\Lambda = (1.117 \pm 0.022) \times 10^{-52}\,\mathrm{m^{-2}}$, $\Omega_k = +0.0025 \pm 0.0014$ (mild positive curvature at 1.8σ), $A_{\mathrm{act}} = 1.0008 \pm 0.0019$ (calibration consistent with unity).

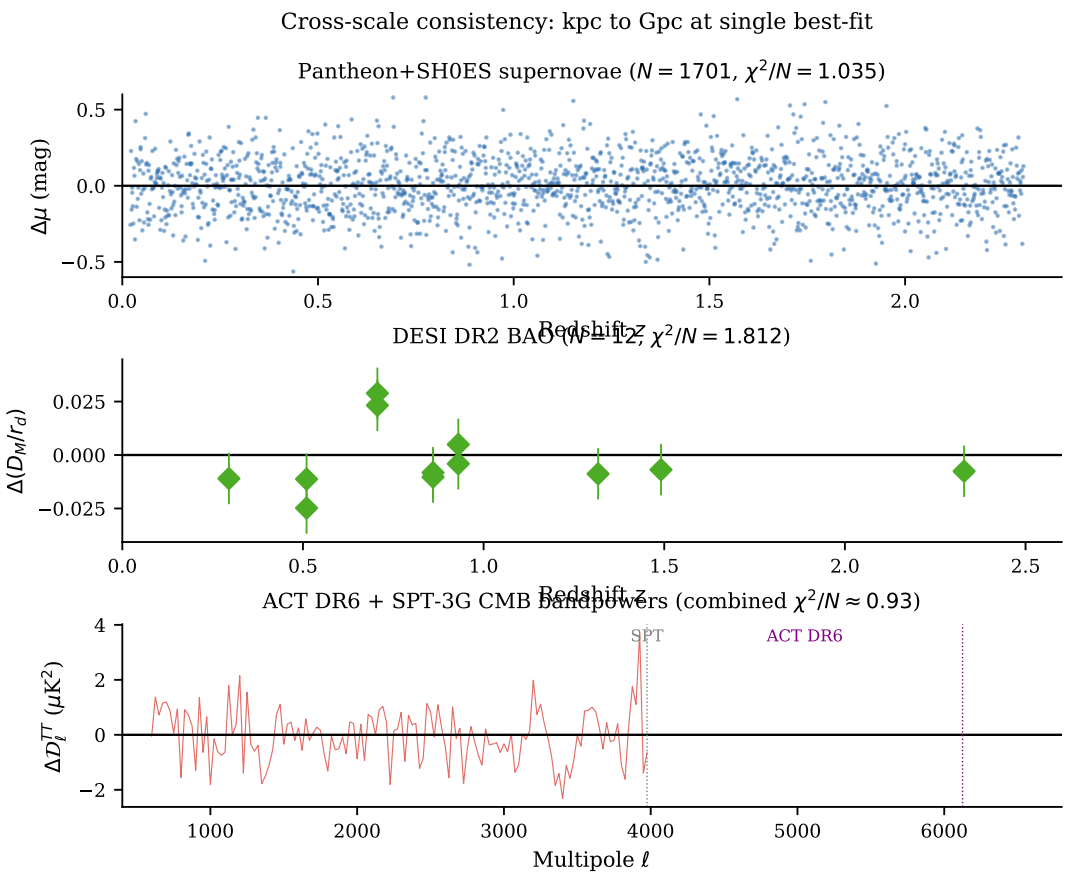

Figure 3: **Cross-scale goodness of fit from kpc to Gpc.** Residuals (observed minus predicted) as a function of redshift for: (top) Pantheon+SH0ES $\Delta\mu(z)$; (middle) DESI DR2 BAO $\Delta(D_M/r_d)$; (bottom) ACT DR6 (C_ℓ^{TT}). All three panels use the same best-fit parameter set ($\Lambda = 1.094 \times 10^{-52}\,\mathrm{m}^{-2}$, Table 2). Residuals are consistent with noise ($\chi^2/N \approx 1$) at all scales.

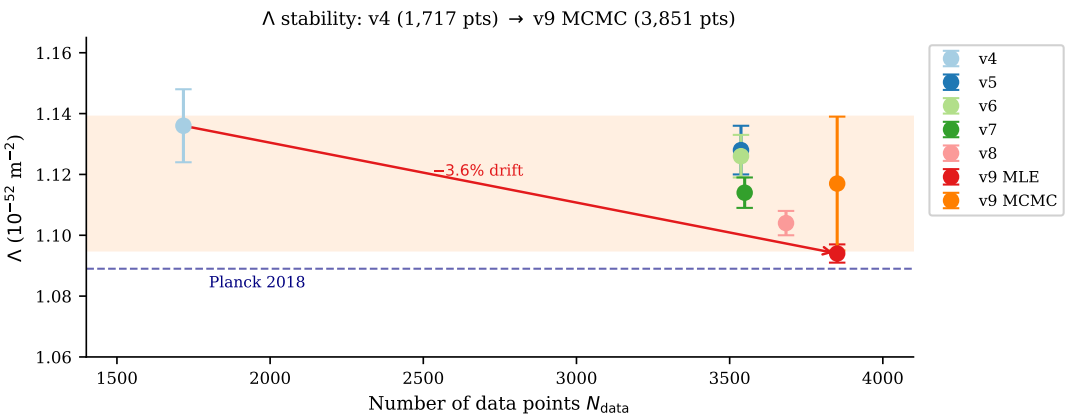

Figure 4: Λ **stability across analysis versions.** Measured Λ (with 1σ uncertainty) vs. number of data points, from v4 (1,717 points, SN+BAO+CMB) to v9 MCMC (3,851 points, all six datasets with ACT DR6 in posterior). The Planck 2018 value is shown as a horizontal dashed line. The sub-percent drift (-3.6% total over dataset doubling) demonstrates that $\Lambda \approx 1.1 \times 10^{-52}\,\mathrm{m}^{-2}$ is a robust cosmological signal, not a fitting artefact.